Rule of 72 (s-RO72.R)

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Description:
This interactive **R** utility estimates compound interest doubling times. It compares the standard **Rule of 72** mental math shortcut against **exact annual discrete compounding** and **continuous compounding limits**, displaying formatted comparative outputs and calculation variances.

Instructions for Use:

1. Prerequisites:
This script runs using base **R** capabilities and requires no external package installations.

2. Execution Steps:

1. Start Interactive Session: Launch the script in **R** console, **RStudio**, or terminal.
Source **s-RO72.R** in RStudio or run it directly in an interactive **R** console:

```
source("s-RO72.R")
```

(Note: You can also execute it non-interactively via command line by passing a rate argument: Rscript s-RO72.R 8)

2. Input Interest Rate: Enter an annual interest percentage.
When prompted by Enter Interest Rate (**%**):, type your target annual rate as a positive number (e.g., 6.5 or 8) and press **Enter**.

3. Review Comparative Results: Analyze compounding metrics and variances.
The console will print an aligned summary comparing the estimation methods:

At a 8.00% annual interest rate:

- **Rule of 72 Estimate: ~9.00 years**
- **Exact (Annual): 9.01 years**
- **Estimation Variance: -0.01 years**
- **Continuous Limit: 8.66 years**

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4. Terminate Session: Type exit command when finished.

To exit the loop, type **q**, quit, exit, or **esc** at the input prompt.